



Fig. 3: Typical life cycle of a wearable device

nents like capacitors and coils have precise dimensions and need to be placed correctly.

Another key component that most designers underestimate is miniature connector. While there is no dearth of miniature connectors, these come with a big restriction on the number of insertions. This is due to connector geometry. In most cases, the number of insertions is in the range of 30 to 40 cycles.

If the assembly process takes more insertion cycles, it can reduce the active life of the connector. (When I was designing a movie camera, we had three to four connectors that were getting cycled 20 to 30 times during the assembly process. The connectors started failing when the final system assembly was done, leading to a complete revamp of the assembly and test process.)

7. Designers should completely understand use cases. If their devices need special chargers or special supplies, they need to plan that as part of the product. With proliferation of USB based charging, most devices tend to supply the product without chargers. However, all chargers are not the same. So if your device is expecting a special treatment, then that has to be provided. If not, the device should be tested for chargers available in the market to ensure there are no surprises.

8. Most wearable devices are waterproof (in technical jargon these are called IP67 compliant). This essentially means that the device enclosure design has to be leak-proof up to a depth of at least 50 metres when submerged in water. This needs skill and experience. The manufacturing process should be able to

reproduce protection for each piece that is manufactured. It is advisable that a relevant subject matter expert is hired as a consultant to review the design and ensure the packaging is

leak-proof.

9. Another challenge that designers face is: How will the software team develop the software if the wearable device design is going to take time to get the prototypes ready?

I suggest that, make the wearable circuit in a large PCB format with bigger components that are used in the design and have test points, so that software developers get the actual target hardware to develop software and do not have to wait for the final miniature version. I have found this approach very useful as it reduces development time considerably despite an additional investment in the hardware.

10. With size being the premier constraint, sometimes volume and shape of the product forces designers to use a complex architecture for the design. Most wearable devices today use a mix of PCBs and flexible circuits to realise the final design. This approach needs a good understanding of these two materials and also how to use these.

The contract manufacturer (electronic manufacturing service or EMS) must be identified as soon as the design is started. Challenge with flexible PCBs is that, every time you revise these, there is substantial upfront expense (non-recurring expense) involved, and multiple revisions can drive up development costs.

Manufacturing phase. The second important stage is the manufacturing phase. A product will be successful only when the design and manufacturing teams work together. When wearable devices are designed, it is important that, at the start of the project, the EMS partner

is decided. Let us see some challenges that wearable products face in this phase.

A conventional SMT process invariably is inadequate for wearable devices due to the following reasons:

1. Miniature size of the product with very less volume and space available inside the enclosure leads to a complex assembly process.
2. In most cases the battery consumes the major volume, leaving little space for the electronics. This leads to very tight packaging and, unless adequate precautions are taken, yields can be a challenge.
3. The smaller PCBs and the use of flex cables sometimes calls for development of special jigs and processes like ultrasonic soldering of flex cables.
4. With reduction in size, many times a product has to be manufactured using packing techniques like chip on board (CoB); normal IC packages and dies that are packaged in standard packages with leads increase the volume manyfold as compared to using the die directly. In a CoB process, the die is directly soldered to the PCB and requires special equipment. Unfortunately, this technique is the only way by which product volume can be reduced substantially.

With component size being the biggest deciding factor, it is important that the designer works with the supply chain of EMS vendors to select the right package size so that, apart from the realisable design, availability of components is also guaranteed. EMS vendors usually have a very good supply chain. They ensure there are no shortages, so that increased lead time does not impact the production schedule.

In the second part of this article, we will discuss the testing phase, wearable system software and development phase.

To be concluded next month...